

# Numerical Integration and Differentiation

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1. Write a C program to evaluate the derivative of  $\sin(x)$  at  $x = \frac{\pi}{4}$  for different values of  $h$ . Plot the results and comment on the importance of  $h$  and differentiation scheme
  - (a) Forward Difference Scheme
  - (b) Backward Difference Scheme
  - (c) Central Difference scheme

2. Write a C program to calculate the area of a circle with  $r = 1$  m without using the value of  $\pi$ 
  - (a) Using Monte Carlo technique
  - (b) Develop an adaptive integration scheme such that any further increase in the number of sectors(or triangles) leads to less than 1% change in the area calculated

$$\frac{A_{old} - A_{new}}{A_{new}} \times 100 \leq 1 \quad (1)$$

3. Given the equilibrium electrostatic potential profile in the diode calculate

- (a) Electric Field  $\mathcal{E} = -\frac{dV}{dx}$
- (b) Charge density  $\rho(x) = \epsilon_{Si} \frac{d\mathcal{E}}{dx}$

4. The electron mobility depends on the electric field along the transport. This dependence is given by

$$\mu = \mu_0 / (1 + \frac{\mathcal{E}}{\mathcal{E}_C})$$

where  $\mu_0$  is the low field mobility ( $= 1400 \text{ cm}^2/V - s$ ) and  $\mathcal{E}_C$  is the critical electric field ( $= 1 \text{ MV/m}$ )  
Considering the 1D diode and the electrostatic potential profile in the diode, calculate position dependent mobility.

5. Calculate the electron,  $n(x)$ , and hole  $p(x)$  concentration in silicon nanowire based FET. In this case, the  $n(x)$  and  $p(x)$  are given by

$$n(x) = \int_{\varepsilon_i}^{\infty} g_{1D}(E) f(E) dE \quad (2)$$

$$p(x) = \int_{-\infty}^{\varepsilon_j} g_{1D}(E) (1 - f(E)) dE \quad (3)$$

$$g_{1D}(E) = \frac{\sqrt{2m_{DOS}}}{\pi\hbar} \frac{1}{\sqrt{E - \varepsilon_i}} \quad (4)$$

Assume  $E_F = 0$ ,  $\varepsilon_i = 0.1 \text{ eV}$  and  $\varepsilon_j = -0.1 \text{ eV}$

6. The wavefunction ( $\zeta(x)$ ) as calculated from the standard solvers is provided in the form of the data. Check if it has been normalized ? If not, then normalize it such that the normalized wavefunction,  $\zeta_N(x)$ , satisfies the following condition

$$|\zeta_N(x)|^2 = 1$$